

Atharva Kirkole

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EDUCATION

Accelerated path: completed B.S. in 3 years + M.S. in 1 year in Computer Science.

Michigan State University

M.S. Computer Science — GPA 3.93 | Teaching Assistant, OOP, Parallel Programming

East Lansing, MI
Aug 2025 – May 2026

Michigan State University

B.S. Computer Science, Minor in Business (Honors College) — GPA 3.93, Honors, Tau Beta Pi

East Lansing, MI
Aug 2022 – Apr 2025

EXPERIENCE

Heard AI

Speech AI Research Assistant

East Lansing, MI
Mar 2026 – Jun 2026

- Contributed to \$5M NSF Convergence Accelerator (Phase 2) project on accessible speech technologies; built automatic speech recognition (ASR), speaker diarization, and NLP dysfluency-classification models to detect and classify dysfluent patterns.
- Built a stutter-aware text-to-speech (TTS) synthesis/regeneration pipeline generating fluent speech from stuttered input via stream and real-time, low-latency inference; serving 300 concurrent sessions at under 400ms latency with live voice feedback.
- Implemented per-user fine-tuning framework achieving 23% WER improvement (14% → 10.8%) using rapid adaptation.

SPaRTa Lab (Scalable Parallel Technologies), MSU

High Performance Computing Research Assistant

East Lansing, MI
Jun 2026 – Present

- Parallelized the linearized Stokes equation over 10M+ sphere- and ellipsoid-surface points, simulating rigid bodies in viscous (low-Reynolds) flow — with applications in red-blood-cell dynamics and drug-discovery modeling.
- Built a CUDA boundary-integral solver reaching 60× speedup via coalesced memory access and shared-memory tiling; applied numerical analysis and error-control techniques to n-body interactions, reducing numerical error from 7% to 2%.

Michigan State University IT Services

Software Developer

East Lansing, MI
Jun 2024 – Apr 2026

- Led end-to-end migration of the campus mapping platform off Google Maps onto a self-managed ESRI ArcGIS stack serving 100K+ daily requests, rebuilding the React/TypeScript map surface around geospatial layers and cutting licensing cost ~20%.
- Owned geospatial data modeling for ~12K+ campus entities/POIs; built automated telemetry ingestion pipelines and REST API processing real-time location signals to drive data-driven route optimization, cutting compute latency ~30% (210ms → ~145ms).
- Engineered student/people information-system REST APIs and Dockerized service images (PHP, JavaScript, TypeScript) via CI/CD, optimized front-end rendering and bundle size and raised site accessibility from 80% to **99.7%** (WCAG 2.1 / ADA / ARIA)

Amazon

Capstone Project

East Lansing, MI
Jan 2025 – May 2025

- Architected an AI-powered semantic code-search system over ~10k internal architecture assets as containerized (Docker) backend microservices exposing REST APIs, backed by AWS OpenSearch, Lambda, DynamoDB, and S3.
- Designed a RAG-style retrieval pipeline (dynamically chunked vector embeddings via Bedrock + SageMaker, indexed in OpenSearch) serving queries at ~90ms; tuned indexing/ranking to lift top-k relevance ~18% (NDCG@10) and added vector-similarity employee recommendations over GitHub signals.

Thompson Maize Lab

Machine Learning Research Intern

East Lansing, MI
Aug 2023 – May 2024

- Built and evaluated Random Forest / ML models (TensorFlow, Python, R) for crop-yield prediction (~72% accuracy), querying 120+ dimensional agronomic datasets from PostgreSQL presented at NAPPN Conference, Purdue, published abstract ([link](#)).
- Developed internal web applications and interactive dashboards (Python, Flask, Django) to visualize experiment results.

PROJECTS

FitTrack — Wearable Time-Series Analytics & Health Prediction (Python / TensorFlow / Big Data)

2025

- Built a real-time fitness analytics platform ingesting high-volume wearable sensor streams (~50Hz) via streaming pipelines with time-series PostgreSQL storage and a live metrics dashboard. Trained ML models predicting age, BMI, and body-fat % from sensor signals — ~88% accuracy, ~3.5% MAE on body-fat regression across ~1.5M samples.

Distributed Systems (MPI) – Masters Parallel Computing Course (4.0)

2026

- Optimized MPI stencil computations using non-blocking communication, RMA (MPI_Win/MPI_Get), custom datatypes, and subcommunicators to scale across 100M+ elements per rank.

HomeApp — Household Inventory & Warranty Tracker (JavaScript / React / Node.js) — [LINK](#)

2025

- Built a full-stack household inventory and warranty-tracking web app in JavaScript backed by a Node.js/Express REST API.
- Implemented JWT/OAuth authentication, image upload workflows, and OCR-based AI pipelines to automatically extract product metadata (brand, model, purchase date, warranty details) from receipts and images; developed a recommendation system to match users with relevant service technicians based on item type, issue category, and location.
- Built dual marketplace flow — a standalone browse and book directory and a reverse marketplace where users describe an issue and providers submit competing, ranked bids.

OceanGPU — GPU-Accelerated Ocean Wave Simulation (CUDA / C++17 / HPC) — [LINK](#)

2026

- Built a HPC ocean/wave simulation engine in CUDA, implementing spectral (FFT-based) numerical methods for hydrodynamics.
- Profiled and benchmarked kernels with Nsight; exposed Python bindings (pybind11) visualization.
- Optimized GPU throughput via shared-memory tiling, coalesced memory access, and CUDA streams overlapping compute with data transfer; added multi-GPU domain decomposition for a 40×+ speedup over a CPU baseline.

TECHNICAL SKILLS

Languages & Frontend: JavaScript, TypeScript, Python, C++, CUDA, SQL, Bash, PHP; React, Angular, Redux, Leaflet.js
Backend & Cloud/DevOps: Node.js, Express.js, Flask, Django, .NET, REST APIs; AWS, Lambda, SageMaker, Docker, Git, CI/CD
AI/HPC & Databases/Maps: PyTorch, CUDA, RAG, MPI, TensorFlow, Hadoop; PostgreSQL, DynamoDB, OpenSearch, ArcGIS